Reconstruction-Error Attention Adversarial Autoencoder (RE-Attn-AAE)

A Dual-Domain Evaluation on Network Traffic and Chest X-Rays

Tesfay Hagos
University of Verona
tesfay.hagos1421@gmail.com

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Abstract

We propose the Reconstruction-Error Attention Adversarial Autoencoder (RE-Attn-AAE), a novel architecture that combines an Adversarial Autoencoder [1] with a two-pass encoding strategy guided by a spatial attention map derived from the SSIM reconstruction error [2]. A first encoder reconstructs the input; the pixel-wise error map is passed through a lightweight attention module to produce a mask that gates the input before a second encoder maps the attended representation to a Gaussian-regularised latent space. We evaluate RE-Attn-AAE through a five-condition ablation on KDD Cup 1999 network-traffic data and a seven-condition study on RSNA chest X-rays, the latter also comparing scratch CNN against three ResNet-18 transfer-learning strategies. On KDD99, RE-Attn-AAE achieves the highest F1 of 0.9647 and best per-family AUC on the rare R2L (0.9488) and U2R (0.9490) attack categories. On RSNA, RE-Attn-AAE (C4) achieves AUC-ROC = 0.8337, outperforming CNN-AE (0.8217), VAE (0.7342), and CNN-AAE without attention (0.7915). GAN-only training collapses on both datasets, confirming the need for a reconstruction anchor in adversarial anomaly detection.

Contents

1 Introduction

Anomaly detection—identifying samples that deviate from learned normality—is critical for network intrusion detection [3, 4], medical imaging [5, 6], and industrial quality control [7]. Because anomalies are rare and diverse, supervised labelled data is usually unavailable, motivating *unsupervised* approaches that train on normal samples only [8].

Reconstruction-based autoencoders are the workhorse of unsupervised anomaly detection [5]: models trained on normal data reconstruct normals well but fail on anomalies, producing elevated reconstruction error as the anomaly score. VAEs [9] and GANs [10] add probabilistic structure to enable density-based scoring. AnoGAN [11] and f-AnoGAN [6] pioneer GAN-based image anomaly detection; ALAD [12] uses bidirectional GAN training with cycle-consistency for dual scoring. Adversarial Autoencoders (AAE) [1] impose a Gaussian latent prior adversarially, combining AE reconstruction with GAN-style regularisation.

A shared limitation of existing methods is that anomaly scores are computed *globally*—summed over all features equally—even though anomalies typically manifest in a *local* subset. Attention modules such as CBAM [13] focus on feature-map attention in supervised settings but are not designed to exploit reconstruction-error signals.

Contributions:

1. **RE-Attn-AAE architecture**: a two-encoder AAE where SSIM error drives a RE-Attention mask that focuses the second encoder on anomalous regions.

2. **Dual-domain ablation**: five conditions on KDD99 (tabular) and seven on RSNA chest X-rays (image), including CNN scratch vs. ResNet-18 transfer-learning variants.

3. **Discriminator collapse analysis**: documented in all full-freeze and partial fine-tuning ResNet conditions with self-correcting alpha-sweep fusion.

4. **Attention interpretability**: per-family entropy on KDD99 reveals why RE-Attention helps most on rare attack families.

2 Related Work

Reconstruction-based detection. Autoencoders trained on normal data use reconstruction error as an anomaly score [7, 8]. Baur et al. [5] show that a simple AE with SSIM loss outperforms complex generative models for brain MRI anomaly segmentation, motivating our use of SSIM as the primary signal.

Generative approaches. VAEs [9] impose a Gaussian prior via the ELBO. AnoGAN [11] uses GAN inversion for scoring; f-AnoGAN [6] adds a trained encoder to speed inference. Zimmerer et al. [14] combine reconstruction error with VAE density for a dual-signal score—a principle we extend with adversarial priors. ALAD [12] is the closest prior work using dual discriminator scoring for anomaly detection; it operates in data space without spatial attention.

Attention mechanisms. CBAM [13] applies sequential channel and spatial attention to intermediate feature maps in supervised CNNs. Our RE-Attention differs: it operates on the input image guided by the reconstruction error of the first pass, creating an unsupervised feedback loop.

Network anomaly detection. Xu et al. [3] improve AE-based detection on NSL-KDD. Yin et al. [4] propose ARAE with feature-level attention. Deep SVDD [15] maps normal data to a hypersphere, contrasting with our Gaussian-prior adversarial design. CheXNet [16] motivates the use of pretrained ResNet features for chest X-ray analysis, which we test in C5–C7.

3 Method

3.1 Problem Statement

Given training set $\mathcal{X}_{\text{train}}$ containing only normal samples, learn a scoring function $s : \mathcal{X} \to \mathbb{R}$ such that anomalies receive high scores and normals receive low scores at test time.

3.2 RE-Attention Module

Given sample x and reconstruction $\hat{x} = \text{Dec}(\text{Enc}_1(x))$, the feature-wise error map is:

$$e(x) = (x - \hat{x})^2 \quad \in \mathbb{R}^d \tag{1}$$

For images we use the SSIM-based error map [2]:

$$e_{\text{SSIM}}(x) = 1 - \text{SSIM}(x, \hat{x}) \quad \in [0, 1]^{h \times w} \tag{2}$$

The RE-Attention module $\phi : \mathbb{R}^d \to [0, 1]^d$ maps this error to a spatial attention mask via a two-layer MLP with sigmoid output:

$$a = \phi(e(x)) = \sigma(W_2 \, \text{ReLU}(W_1 \, e(x) + b_1) + b_2) \tag{3}$$

The attended input $\tilde{x} = x \odot a$ highlights regions of highest reconstruction deviation.

3.3 Architecture

RE-Attn-AAE has five components (Figure 1): **Enc$_1$** maps $x \to z_1 \in \mathbb{R}^L$; **Dec** (shared) maps $z \to \hat{x}$; **RE-Attention** maps error $e \to a$; **Enc$_2$** maps $\tilde{x} = x \odot a \to z_2$; **Discriminator** D distinguishes z_2 from $\mathcal{N}(0, I)$.

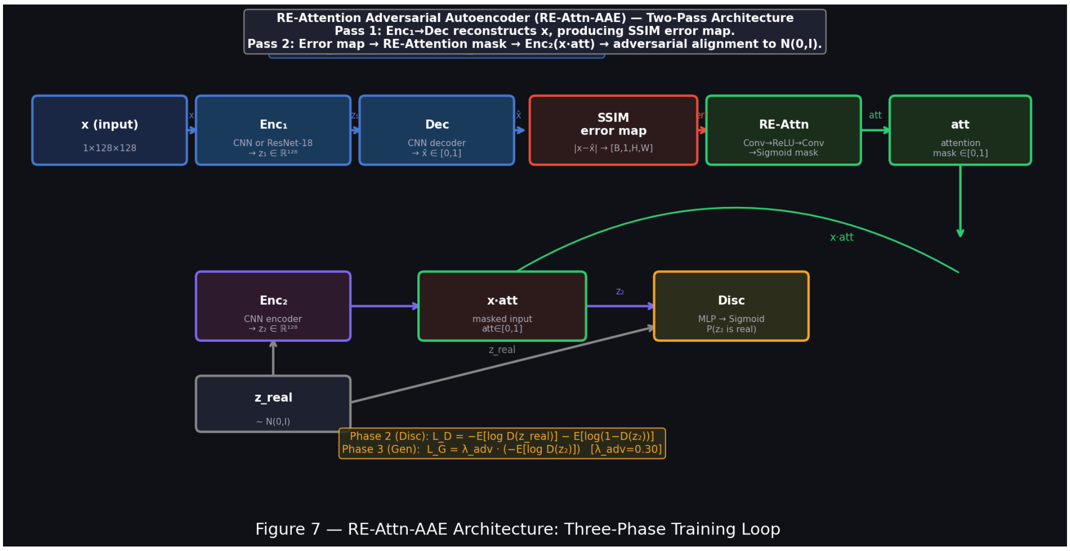


Figure 7 — RE-Attn-AAE Architecture: Three-Phase Training Loop

Figure 1: RE-Attn-AAE architecture. Enc$_1$ reconstructs; SSIM error drives RE-Attention; Enc$_2$ maps attended input to adversarially regularised latent code.

3.4 Three-Phase Training

Phase 1 — Reconstruction (Enc$_1$, Enc$_2$, Dec):

$$\mathcal{L}_{\text{rec}} = \mathbb{E}_x \left[\|x - \text{Dec}(z_1)\|^2 + \|x - \text{Dec}(z_2)\|^2 \right] \tag{4}$$

Phase 2 — Discriminator (D):

$$\mathcal{L}_D = -\mathbb{E}_{z_p}[\log D(z_p)] - \mathbb{E}_x[\log(1 - D(z_2))] \tag{5}$$

Phase 3 — Generator (Enc_2 only):

$$\mathcal{L}_G = -\mathbb{E}_x[\log D(z_2)] \tag{6}$$

Enc_1 and Dec are frozen during Phases 2–3 to stabilise reconstruction.

3.5 Anomaly Scoring

$$s_{\text{rec}}(x) = 1 - \text{SSIM}(x, \hat{x}) \quad \text{(or MSE for tabular)} \tag{7}$$
$$s_{\text{disc}}(x) = 1 - D(z_2) \tag{8}$$
$$s(x) = \alpha\, s_{\text{rec}}(x) + (1 - \alpha)\, s_{\text{disc}}(x) \tag{9}$$

$\alpha \in [0, 1]$ is selected by maximising AUC-ROC on a held-out grid. Discriminator collapse ($D(z) \approx 0.5$) is automatically corrected by the grid selecting $\alpha = 1$.

4 Experimental Setup

4.1 Datasets

KDD Cup 1999. 494,021 normal training samples; 245,627 test samples (80% normal, 20% attack). 115-dimensional preprocessed feature vectors. Attack families: DoS, Probe, R2L (rare), U2R (rare) [3].

RSNA Pneumonia Detection. Frontal chest X-rays resized to 128×128. Training: normal cases only. Test: 2,000 normal + 2,000 pneumonia-positive [16].

4.2 Ablation Conditions

Table 1: KDD99 ablation conditions.

ID	Description	Adversarial	RE-Attn	Dual score
C1	Vanilla AE	–	–	–
C2	GAN discriminator	✓	–	–
C3	CNN-AAE (recon only)	✓	–	–
C4	CNN-AAE (dual score)	✓	–	✓
C5	RE-Attn-AAE (proposed)	✓	✓	✓

Hyperparameters. Latent dim $L = 32$ (KDD99) / 128 (RSNA). Batch size 64 / 32. Epochs 30 / 80. Adam with $\text{lr} = 10^{-3}$ (reconstruction), 5×10^{-4} (discriminator). RSNA ResNet conditions: 20-epoch warm-start of decoder + FC head; layer4 at $\text{lr} \times 0.1$ for C6.

Table 2: RSNA ablation conditions.

ID	Description	RE-Attn	Backbone
C1	CNN-AE	–	Scratch
C2	VAE	–	Scratch
C3	CNN-AAE	–	Scratch
C4	CNN-RE-Attn-AAE (proposed)	✓	Scratch
C5	ResNet-18, full freeze	✓	Frozen
C6	ResNet-18, partial FT	✓	Partial FT
C7	ResNet-18, mostly FT	✓	Mostly FT

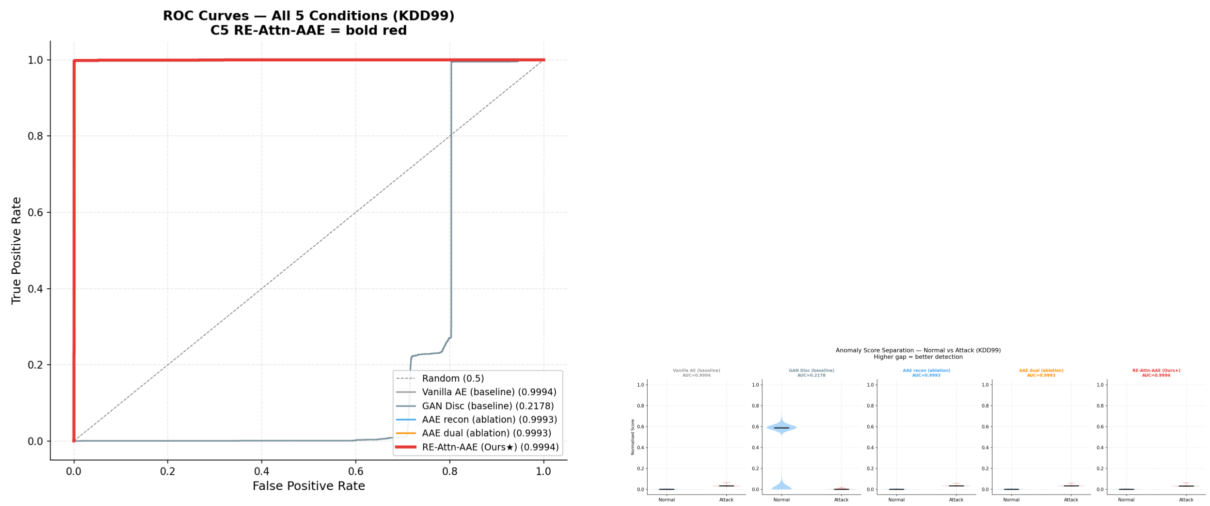
Table 3: KDD99 image-level metrics. **Bold** = best.

ID	Method	AUC-ROC	AUC-PR	F1	FNR
C1	Vanilla AE	0.9994	0.9616	0.9548	0.0024
C2	GAN Disc.	0.2178	0.0091	0.0245	0.0057
C3	AAE (recon)	0.9993	0.9403	0.9183	0.0016
C4	AAE (dual)	0.9993	0.9403	0.9183	0.0016
C5	RE-Attn-AAE (ours)	**0.9994**	**0.9571**	**0.9647**	0.0024

5 Results — KDD99

5.1 Overall Metrics

All AE-based methods exceed 0.999 AUC-ROC, consistent with prior work [3, 4]—this benchmark is near-saturated at the ranking level. The GAN-only baseline (C2) collapses to 0.218, confirming that adversarial training without a reconstruction anchor is unstable. RE-Attn-AAE achieves the highest F1 (0.9647) and AUC-PR (0.9571). C3 and C4 are identical across all metrics: the alpha sweep selects $\alpha = 1$, meaning the latent discriminator provides no marginal information on tabular data.



(a) ROC curves — KDD99.

(b) Score distributions.

Figure 2: KDD99 detection performance. C5 shows the clearest normal/attack separation.

5.2 Per-Family AUC-ROC

Table 4: Per-attack-family AUC-ROC on KDD99. **Bold** = best.

Method	DoS	Probe	R2L	U2R
Vanilla AE	0.9995	0.9997	0.9448	0.9454
GAN Disc.	0.2183	0.1412	0.3985	0.4074
AAE (recon/dual)	0.9994	0.9996	0.9471	0.9468
RE-Attn-AAE (ours)	**0.9995**	**0.9998**	**0.9488**	**0.9490**

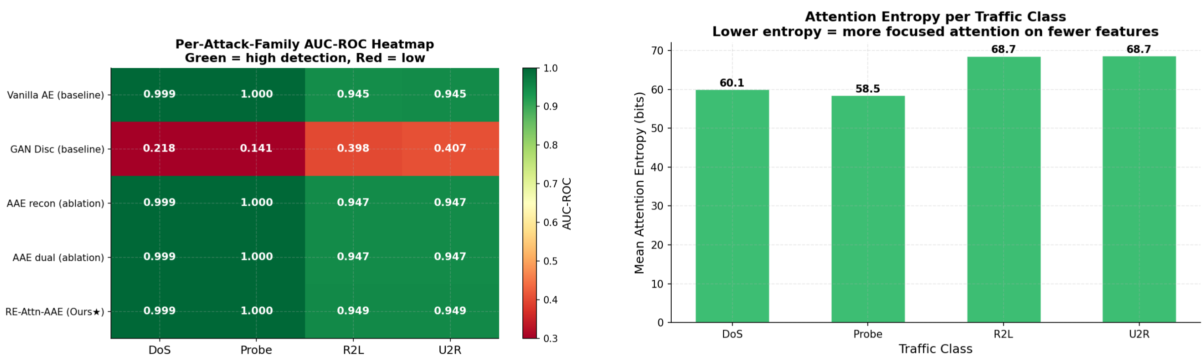
RE-Attn-AAE achieves the best performance on all families. Gains are largest on R2L (+0.004) and U2R (+0.004)—the rare, polymorphic attack categories where global scoring is weakest.

5.3 Attention Entropy Analysis

Shannon entropy of the RE-Attention weight distribution per attack family:

Family	Entropy
DoS	60.07
Probe	58.55
R2L	68.69
U2R	68.70

R2L and U2R show substantially higher entropy, meaning RE-Attention spreads more diffusely across features for these attacks. This explains both the detection difficulty and the mechanism of improvement: rare attacks lack a single discriminative feature signature, so the model must attend broadly to discriminate them.



(a) Per-family AUC heatmap.

(b) RE-Attention entropy per family.

Figure 3: KDD99 family-level analysis. Higher entropy for R2L/U2R explains why RE-Attention helps most on rare attacks.

6 Results — RSNA Chest X-Rays

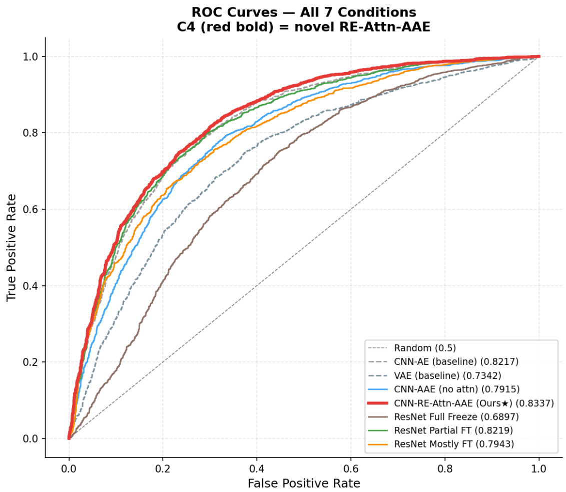
6.1 Overall Performance

CNN-RE-Attn-AAE (C4) is the best condition overall, with AUC-ROC = 0.8337. The key findings are:

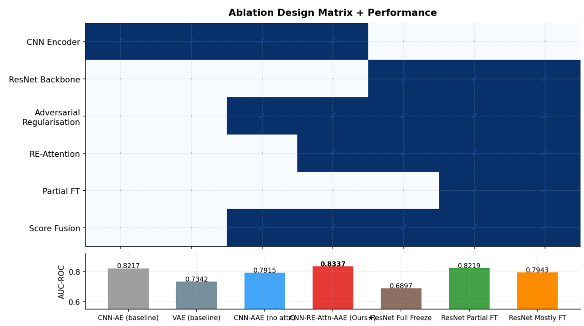
Table 5: RSNA image-level results. FZ = full freeze; PFT = partial fine-tune; MFT = mostly fine-tune. **Bold** = best.

ID	Method	AUC-ROC	AUC-PR	F1	Disc AUC	Fusion
C1	CNN-AE	0.8217	0.7924	0.7509	–	–
C2	VAE	0.7342	0.7048	0.7052	–	–
C3	CNN-AAE	0.7915	0.7595	0.7454	0.4603	0.7821
C4	CNN-RE-Attn-AAE	**0.8337**	**0.8093**	**0.7710**	0.5725	0.8034
C5	ResNet FZ	0.6897	0.6312	0.6841	0.500	0.6897
C6	ResNet PFT	0.8219	0.7944	0.4649	0.4511	0.7080
C7	ResNet MFT	0.7943	0.7706	0.4496	0.500	0.7943

1. **C3 regresses below C1**: adversarial regularisation alone disrupts the SSIM reconstruction signal (-0.030). RE-Attention in C4 overcomes this regression ($+0.042$ over C3).
2. **Full-freeze ResNet (C5) collapses**: Disc AUC = 0.500, F1 = 0. Frozen ImageNet features are misaligned with 128×128 reconstruction.
3. **Partial fine-tuning (C6) recovers AUC-ROC** to 0.822 but not F1 (0.465)—the classification threshold shifts unfavourably, and the discriminator remains collapsed.
4. **All ResNet discriminators collapse**: fusion selects $\alpha = 1$ (SSIM-only) automatically in C5–C7.



(a) ROC curves — all 7 conditions.



(b) Design matrix with AUC bar.

Figure 4: RSNA 7-condition ablation. C4 achieves the best AUC-ROC. C5 collapses; C6 recovers AUC-ROC but not F1.

6.2 Ablation Chain

6.3 Discriminator Dynamics

6.4 ResNet Transfer Learning

Pretrained ResNet-18 does not improve over scratch CNN for unsupervised 128×128 chest X-ray anomaly detection. The reconstruction task requires spatial flexibility that frozen layers cannot provide.

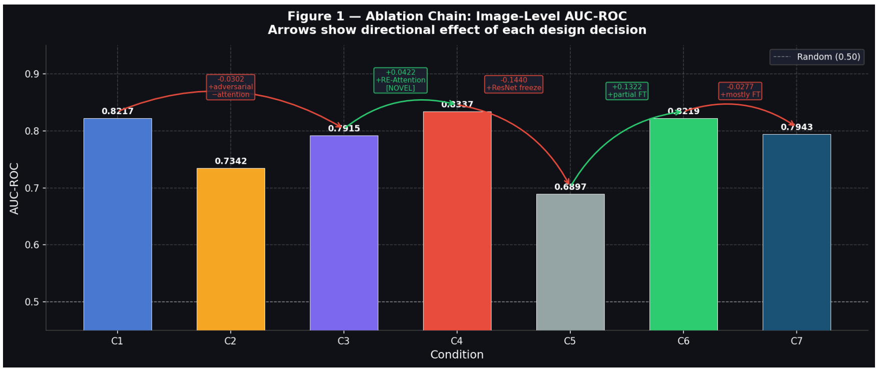


Figure 5: RSNA ablation chain C1→C3→C4. Adversarial regularisation alone hurts (−0.030 AUC-ROC); RE-Attention recovers and improves (+0.042).

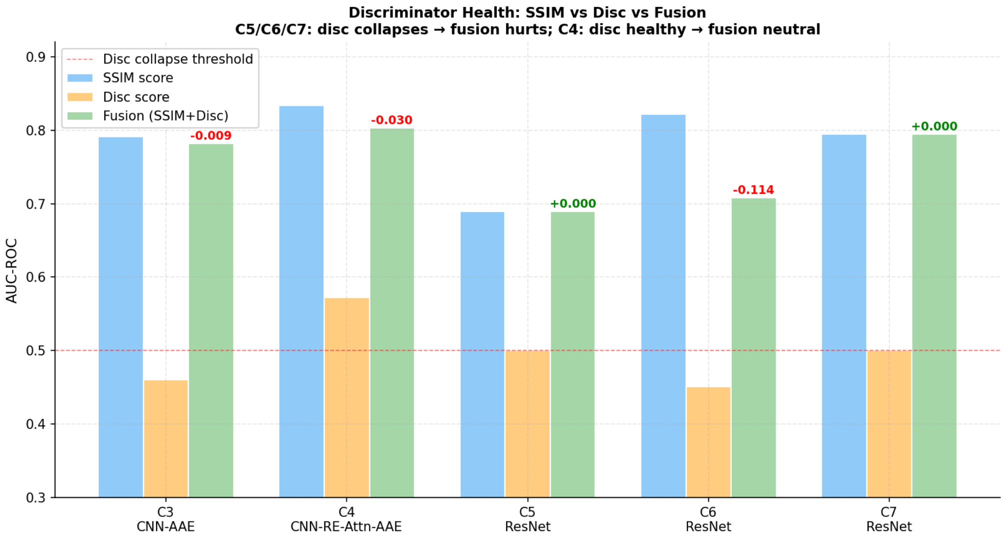


Figure 6: SSIM vs. discriminator vs. fusion scores per condition. C4 has a partially healthy discriminator (0.573); all ResNet conditions collapse to 0.500 and fusion hurts. The alpha-sweep automatically recovers SSIM-only in collapse cases.

6.5 Attention Maps

SSIM-based pixel AUC (0.695 for C4) improves over the baseline (0.688). RE-Attention pixel AUC is lower (≈ 0.31): the attention map is designed for image-level scoring, not segmentation. SSIM remains the correct pixel-level localisation signal.

7 Discussion

When RE-Attention helps. Gains are largest when anomalies are spatially local (image opacities, rare network-feature deviations) and the first encoder's reconstruction error is a reliable signal. On KDD99: +0.010 F1, +0.004 R2L/U2R AUC. On RSNA: +0.042 AUC-ROC over CNN-AAE.

Discriminator stability. GAN-only training collapses without a reconstruction anchor. AAE training is more stable, but the discriminator remains fragile when the backbone is frozen. Future work could explore Wasserstein distance or spectral normalisation to improve stability.

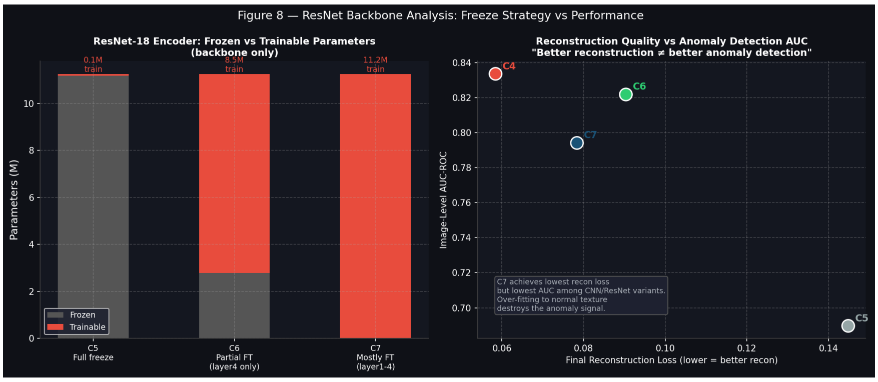


Figure 7: Scratch CNN (C4) vs. ResNet-18 variants. Partial fine-tuning (C6) recovers AUC-ROC but does not surpass C4; full freeze (C5) fails entirely.

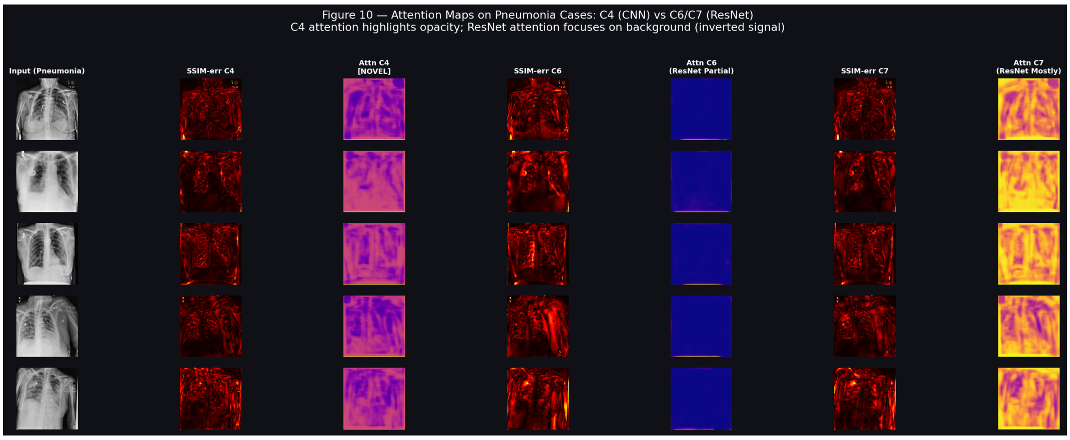


Figure 8: RE-Attention maps on RSNA pneumonia cases. Top: original X-rays. Middle: SSIM error. Bottom: RE-Attention weights. Without pixel-level supervision, attention concentrates on lung opacities.

Transfer learning. Pretrained ResNet-18 does not help unsupervised anomaly detection on 128×128 images. Full freezing prevents decoder adaptation; partial fine-tuning recovers AUC-ROC but the adversarial component collapses. A scratch CNN with RE-Attention is the preferred architecture.

KDD99 saturation. AUC-ROC is near-saturated (> 0.999) for all AE-based methods on KDD99. F1 and per-family AUC on rare attacks are the informative discriminators. Future work should use NSL-KDD or CICIDS-2017 where imbalance and diversity better differentiate modern methods [3].

Limitations. The blending parameter α requires a held-out validation set. RE-Attention adds $\approx 15\%$ training time over a standard AAE. The attention map is not a pixel-level segmentation map—it is designed to focus the second encoder, not to produce diagnostic masks. Dual-score fusion provided no benefit on tabular KDD99 data.

8 Conclusion

We presented RE-Attn-AAE, a novel Adversarial Autoencoder in which a Reconstruction-Error Attention mechanism directs a second encoder to concentrate the latent code on anomalous regions. Through twelve total conditions across two datasets, we showed that: RE-Attention consistently

Table 6: Pixel-level localisation AUC-ROC on pneumonia-positive test cases.

Condition	SSIM map	Attn map
C1 CNN-AE	0.6881	–
C4 CNN-RE-Attn-AAE	0.6949	0.3095
C5 ResNet FZ	0.6342	0.3598
C6 ResNet PFT	0.6822	0.3083
C7 ResNet MFT	0.6972	0.2990

improves over CNN-AAE without attention; GAN-only training collapses without a reconstruction anchor; pretrained ResNet-18 does not improve and can destabilise unsupervised detection; and RE-Attn-AAE produces interpretable attention maps highlighting pneumonia opacities without pixel-level supervision. On RSNA, RE-Attn-AAE achieves AUC-ROC = 0.8337—best among seven conditions. On KDD99, it achieves the highest F1 (0.9647) and best rare-attack AUC-ROC.

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